Outlook

Closure Challenge submission — Shuhan Yang

From yang han <yshcloud@outlook.com>
Date Tue 7/28/2026 11:23 AM
To Ryley McConkey <rmcconke@mit.edu>

1 attachment (123 KB)
closure_submission.zip;

Dear Dr. McConkey,

Attached is a submission to the Closure Challenge: eight prediction CSVs (velocity vectors at the evaluation points, in the order returned by closure_challenge.evaluation_points), a manifest, and a method note.

Self-evaluated with closure-challenge v0.3.1, the overall score is 0.05800:

alpha_15_13929_4048 = 0.08060
alpha_15_13929_2024 = 0.12286
alpha_05_4071_4048 = 0.06449
alpha_05_4071_2024 = 0.07480
AR_1_Ret_360 = 0.02908
AR_3_Ret_360 = 0.03108
AR_14_Ret_180 = 0.02502
NASA_2DWMH = 0.03607

Method, in short: a model-consistent (a-posteriori) algebraic stress correction for k-omega SST, applied to OpenFOAM 7 as a runtime fvOptions source. The anisotropy is

b = sigma(x) [g0 b_bouss + sum_n g_n That_n + g_w dev(n n)]

with That_n the first three Pope basis tensors, n the wall-distance gradient direction, the coefficients linear in eight bounded local invariants, and sigma a shrinkage gate on P/epsilon. The wall-orientation dyad dev(n n) is the one non-standard ingredient. In a fully-developed duct the velocity-gradient algebra closes exactly (W^2 = -S^2), so the ten-term integrity basis collapses onto span{T1,T2,T3}, and the only cross-plane direction it can still produce scales with shear squared — which vanishes precisely in the corners where the secondary vortices live. The dyad is independent of that algebra and stays O(1) there. The note gives the argument in full, together with the two extra features the same degeneracy forced.

The 40 coefficients were fitted with CMA-ES on the propagated velocity error over 24 training flows (19 parametric hills, four ducts at Re_tau 180, the curved backward-facing step), with each physics class weighted by its share of the test set and an extra charge for degrading any flow whose baseline is already accurate. I did not fit frozen stress targets, and the reason may be of interest for the benchmark's framing: imposing the exact high-fidelity Reynolds stress in the momentum equation makes the propagated velocity worse than the unmodified baseline on this dataset — +18% to +73% on three training flows, fixed-point converged and mask-independent — which is consistent with the ill-conditioning reported by Wu et al. (JFM 2019). Happy to share those numbers if they are useful to you.

Two points I should flag.

First, the numerics. Because the correction is an explicit source rebuilt from the evolving field at every momentum solve, the corrected runs plateau near 1e-3 initial residual instead of reaching a machinezero steady state (checked out to 20000 sweeps, with no negative-k bounding anywhere). The metric does settle into a narrow band: sampled at eight points from 2500 to 20000 sweeps the overall score stays within 0.05875-0.05788. Rather than report an arbitrary point in that band, the submitted field is the mean of the solution over the last five snapshots — a rule fixed before its value was known, and slightly worse than the best single snapshot (0.05788). The band is not a closed cycle; it is still drifting slightly downward at 20000 sweeps, so a longer run would most likely score marginally better; I did not pursue it further. Happy to supply the trajectory, or a field defined by whatever convention you would prefer.

Second, test-set access. The configuration was chosen on your suggested validation split plus two training-pool flows at exactly the test operating points. The test cases were scored at four pre-registered milestones — an unmodified baseline and three model milestones — each written to an append-only log, with the baseline re-scored as a control every time. After the final milestone the frozen model was scored at eight further points along its convergence trajectory: that is the band above, and those readings changed no coefficient and no configuration, only which field to submit, under the rule fixed in advance. The method note sets all of this out, so that the log and the submitted number can be reconciled. No test-case data of any kind entered the fit or the model selection.

Two of the eight cells are worse than the unmodified baseline, alpha_05_4071_4048 and, marginally, alpha_05_4071_2024. The method note says so plainly, and observes that on those two cells the plain k-omega SST baseline also beats every entry currently on the leaderboard — it looks like a property of the low-Reynolds, low-alpha hills rather than of any one correction.

For the leaderboard entry:

Authors: Shuhan Yang
Reference: none

Best regards,
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